

[BUG] Update validation check in ``qml.Hermitian`` to support multiple inter

<https://github.com/PennyLaneAI/pennylane/issues/2132>

ROW 108

[BUG] Update validation check in `qml.Hermitian` to support multiple interfaces #2132

New issue



Closed



antalszava opened on Jan 27, 2022

...

Expected behavior

The following executes correctly:

```
import pennylane as qml
import numpy as np
import tensorflow as tf

dev = qml.device('default.qubit', wires=1, shots=10)

arr = np.array([
    1.32,
    2.312,
])

herm = tf.Variable(np.outer(arr, arr))

@qml.qnode(dev, interface='tf')
def circuit():
    return qml.sample(qml.Hermitian(herm, wires=0))

res = circuit()
```



Actual behavior

AttributeError: 'tensorflow.python.framework.ops.EagerTensor' object has no attribute 'conj' is raised due to the

```
if not qml.math.allclose(A, A.conj().T):
    raise ValueError("Observable must be Hermitian.")
```



validation check.

Additional information

To resolve the issue, it may be good to consider if we could support `conj` and/or `T` in `qml.math` for multiple interfaces.

On top of that, however, `qml.math.allclose` will break jittability, hence potentially omitting the validation as a whole should also be considered.

Source code

No response

Tracebacks

```
-----
AttributeError                                Traceback (most recent call last)
<ipython-input-1-0a0e7a84a095> in <module>
    21 print(res, type(res), res.dtype)
    22
--> 23 circuit.qtape.get_output_domain()

~/xanadu/pennylane/pennylane/tape/tape.py in get_output_domain(self)
   1079
   1080         output_domain = (
-> 1081             int if all(np.issubdtype(e.dtype, int) for e in observable.eigvals) else float
   1082         )
   1083

~/xanadu/pennylane/pennylane/measure.py in eigvals(self)
   144         if self.obs is not None:
   145             try:
-> 146                 return self.obs.eigvals
   147             except NotImplementedError:
```



Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



Zot

```

148         pass

~/xanadu/pennylane/pennylane/ops/qubit/observables.py in eigvals(self)
102         array: array containing the eigenvalues of the Hermitian observable
103         """
--> 104         return self.eigendecomposition["eigval"]
105
106     def diagonalizing_gates(self):

~/xanadu/pennylane/pennylane/ops/qubit/observables.py in eigendecomposition(self)
83         dict[str, array]: dictionary containing the eigenvalues and the eigenvectors of the Hermit
84         """
----> 85         Hmat = self.matrix
86         Hmat = qml.math.to_numpy(Hmat)
87         Hkey = tuple(Hmat.flatten().tolist())

~/xanadu/pennylane/pennylane/operation.py in matrix(self)
342         array: matrix representation
343         """
--> 344         return self._matrix(*self.parameters)
345
346     @classmethod

~/xanadu/pennylane/pennylane/ops/qubit/observables.py in _matrix(cls, *params)
66         raise ValueError("Observable must be a square matrix.")
67
----> 68         if not qml.math.allclose(A, A.conj().T):
69             raise ValueError("Observable must be Hermitian.")
70

~/anaconda3/lib/python3.8/site-packages/tensorflow/python/framework/ops.py in __getattr__(self, name)
440         from tensorflow.python.ops.numpy_ops import np_config
441         np_config.enable_numpy_behavior()"".format(type(self).__name__, name))
--> 442         self.__getattrattribute__(name)
443
444     @staticmethod

AttributeError: 'tensorflow.python.framework.ops.EagerTensor' object has no attribute 'conj'

```

System information

Name: PennyLane
Version: 0.21.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/XanaduAI/pennylane>
Author:
Author-email:
License: Apache License 2.0
Location: /home/antal/xanadu/pennylane
Requires: numpy, scipy, networkx, retworkx, autograd, toml, appdirs, semantic_version, autoray, cachetools, psutil
Required-by: PennyLane-Cirq, amazon-braket-pennylane-plugin, PennyLane-Orquestra, pennylane-qulacs, PennyLane-Platform info: Linux-5.13.0-27-generic-x86_64-with-glibc2.10
Python version: 3.8.5
Numpy version: 1.19.2
Scipy version: 1.7.3
Installed devices:
- cirq.mixedsimulator (PennyLane-Cirq-0.19.0)
- cirq.pasqal (PennyLane-Cirq-0.19.0)
- cirq.qsim (PennyLane-Cirq-0.19.0)
- cirq.qsimh (PennyLane-Cirq-0.19.0)
- cirq.simulator (PennyLane-Cirq-0.19.0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.5.3)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.5.3)
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.18.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.18.0)
- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
- projectq.ibm (PennyLane-PQ-0.18.0.dev0)
- projectq.simulator (PennyLane-PQ-0.18.0.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.18.0.dev0)
- forest.qvm (PennyLane-Forest-0.18.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.18.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.19.0)

```

- braket.aws.qubit (amazon-braket-pennylane-plugin-1.5.3)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.5.3)
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.basicq (PennyLane-qiskit-0.18.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.18.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.18.0)
- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
- projectq.ibm (PennyLane-PQ-0.18.0.dev0)
- projectq.simulator (PennyLane-PQ-0.18.0.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.18.0.dev0)
- forest.qvm (PennyLane-Forest-0.18.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.18.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.19.0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- strawberryfields.fock (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.20.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.20.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.20.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.21.0.dev0)
- default.gaussian (PennyLane-0.21.0.dev0)
- default.mixed (PennyLane-0.21.0.dev0)
- default.qubit (PennyLane-0.21.0.dev0)
- default.qubit.autograd (PennyLane-0.21.0.dev0)
- default.qubit.jax (PennyLane-0.21.0.dev0)
- default.qubit.tf (PennyLane-0.21.0.dev0)
- default.qubit.torch (PennyLane-0.21.0.dev0)

```

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



antalszava added bug on Jan 27, 2022



timmysilv on Feb 13, 2023

Contributor ...

duplicate of [#2892](#), resolved



timmysilv closed this as completed on Feb 13, 2023

Update the Hermiticity check of `qml.Hermitian` to use `qml.math` #2895

[Code](#)

Merged antalszava merged 6 commits into `v0.25.0-rc0` from `fix_2892` on Aug 8, 2022

Conversation 6

Commits 6

Checks 0

Files changed 3

+26 -3



antalszava commented on Aug 5, 2022

Contributor

Reworks the Hermiticity check in `qml.Hermitian` by using `qml.math` because calling `.conj()` on an `EagerTensor` from TF raises an error.

Closes #2892



fix

b26d074



github-actions bot commented on Aug 5, 2022

Contributor

Hello. You may have forgotten to update the changelog!
Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



antalszava added 2 commits 3 years ago

changelog

ea5c1c1

lint

de717ea

antalszava requested a review from Jaybsoni 3 years ago



antalszava commented on Aug 5, 2022

View reviewed changes

pennylane/ops/qubit/observables.py

Show resolved



Jaybsoni approved these changes on Aug 5, 2022

View reviewed changes

Jaybsoni left a comment

Contributor

Looks good to me! I am surprised that tf can't natively resolve this, but autoray can!



Reviewers

Jaybsoni



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[\[BUG\] Hermitian operator fails during expva...](#)

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2 participants

